

The Biological Learning Rules I

Quan Wen May 18

The Credit Assignment Problem

The Credit Assignment Problem

When a network produces an error, which synapses (among potentially billions) are responsible?

Spatial Credit

Which neurons and layers contributed to the output error?

Temporal Credit

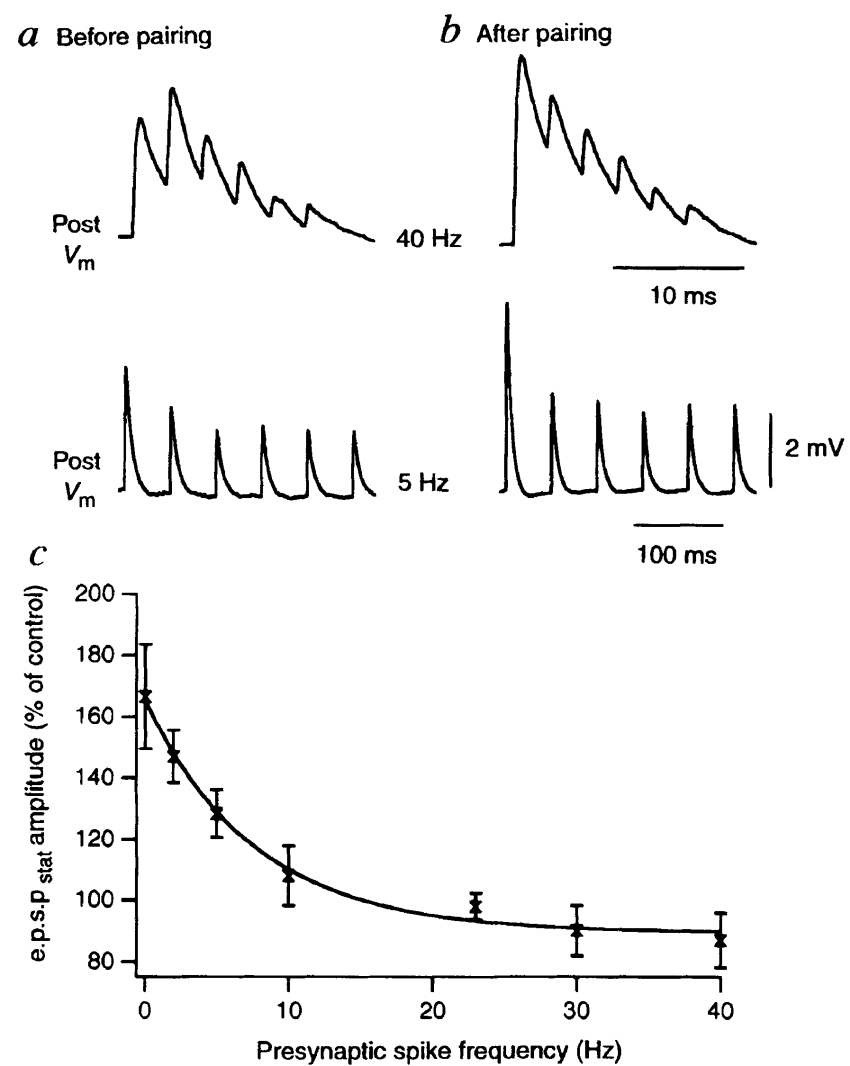
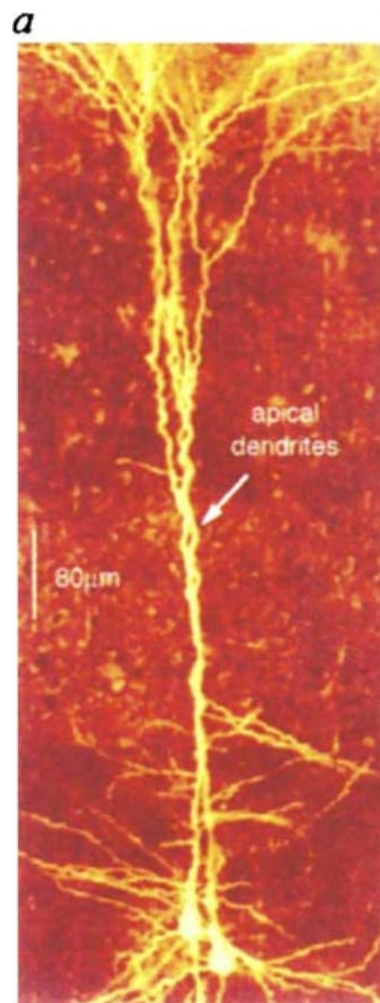
Which past activations (across time steps) are relevant?

Structural Credit

How should each synapse's weight change to reduce error?

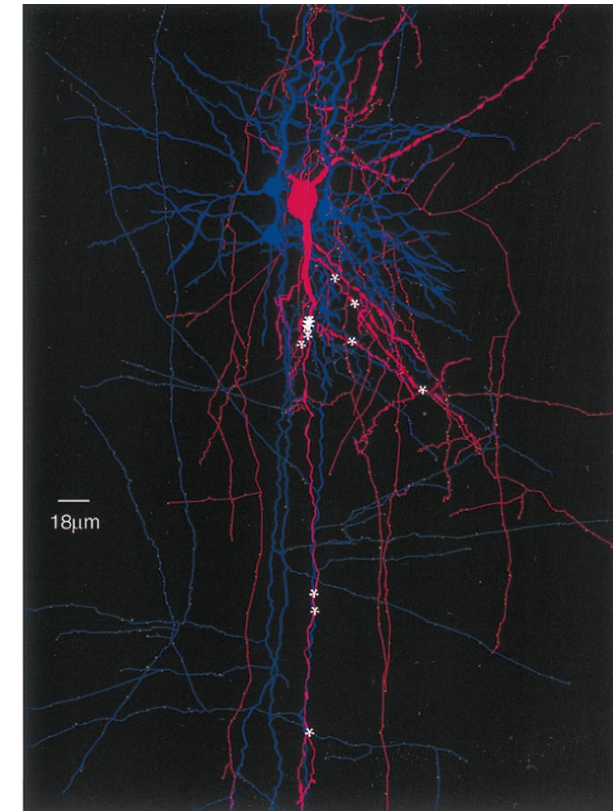
Synaptic plasticity (phenomenon)

Short-term synaptic plasticity

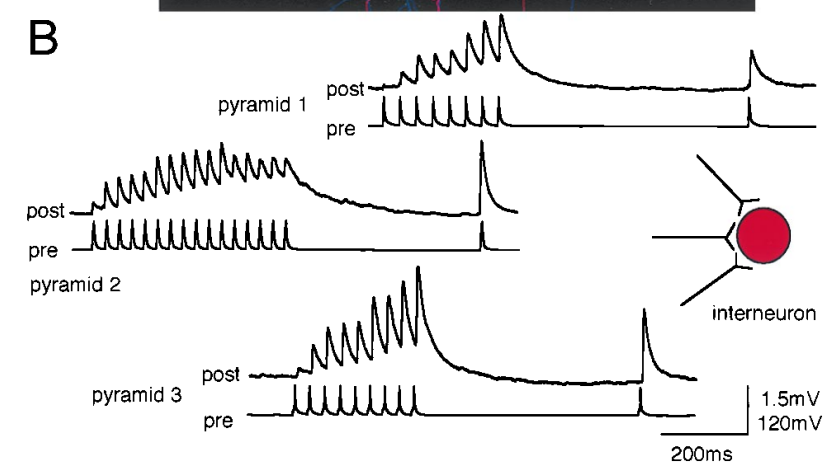


Depression

A



B



Facilitation

A Phenomenological Approach to Dynamic Synaptic Transmission

4 Key Synaptic Parameters

Absolute strength (A)

Probability of release (U)

Depression time constant (τ_d)

Facilitation time constant (τ_f)

$$\frac{du}{dt} = -\frac{u}{\tau_f} + U(1 - u)\delta(t - t_{spike})$$

2 Synaptic Variables

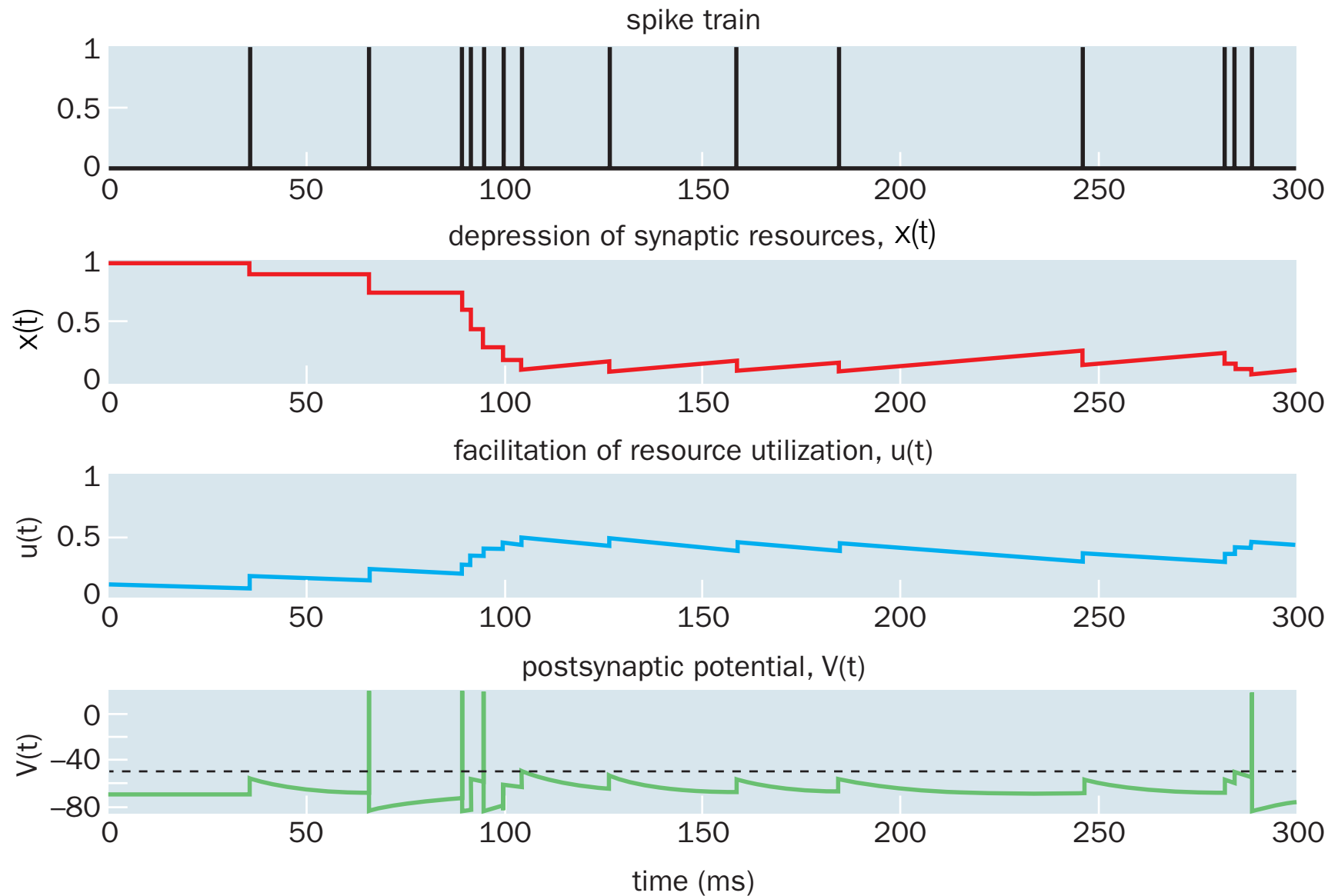
Resources available (x)

Release probability (u)

$$\frac{dx}{dt} = -\frac{1 - x}{\tau_d} - ux\delta(t - t_{spike})$$

$$V(t) = Aux$$

A Phenomenological Approach to Dynamic Synaptic Transmission



$$\frac{du}{dt} = -\frac{u}{\tau_f} + U(1 - u)\delta(t - t_{spike})$$

$$\frac{dx}{dt} = -\frac{1 - x}{\tau_d} - ux\delta(t - t_{spike})$$

$$V(t) = Aux$$

Long-term synaptic plasticity

Hebbian Learning

"Neurons that fire together, wire together"

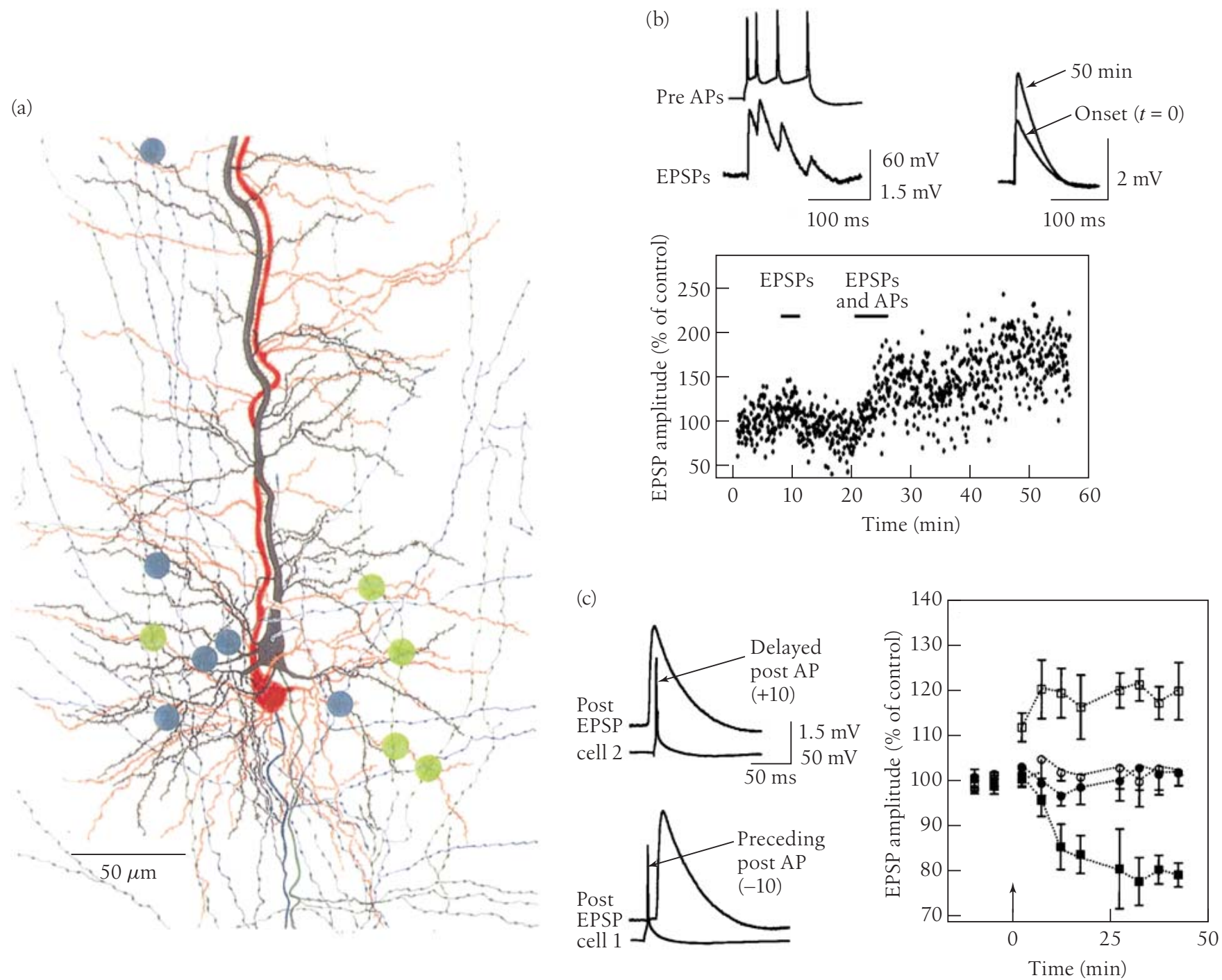
— Paraphrase of Hebb (1949)

$$\Delta w_{ij} = \eta x_i x_j$$

Δw_{ij}	Change in synaptic weight
η	Learning rate
x_i, x_j	Pre- and post-synaptic activity

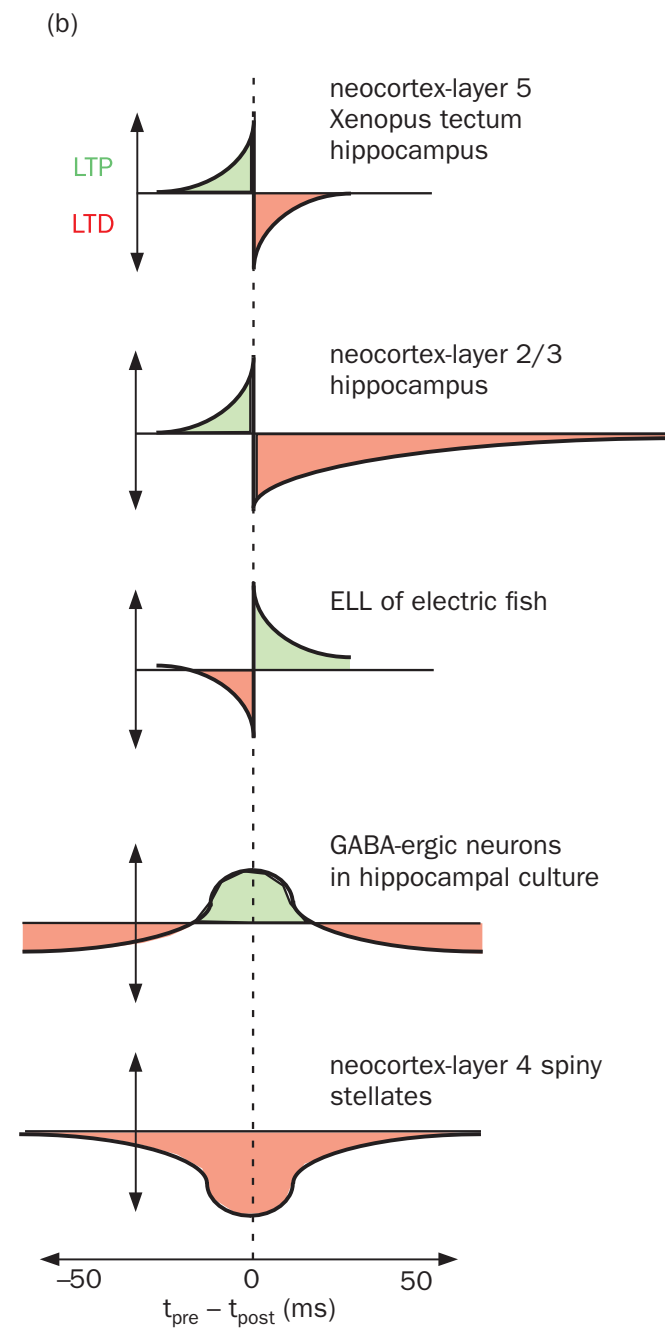
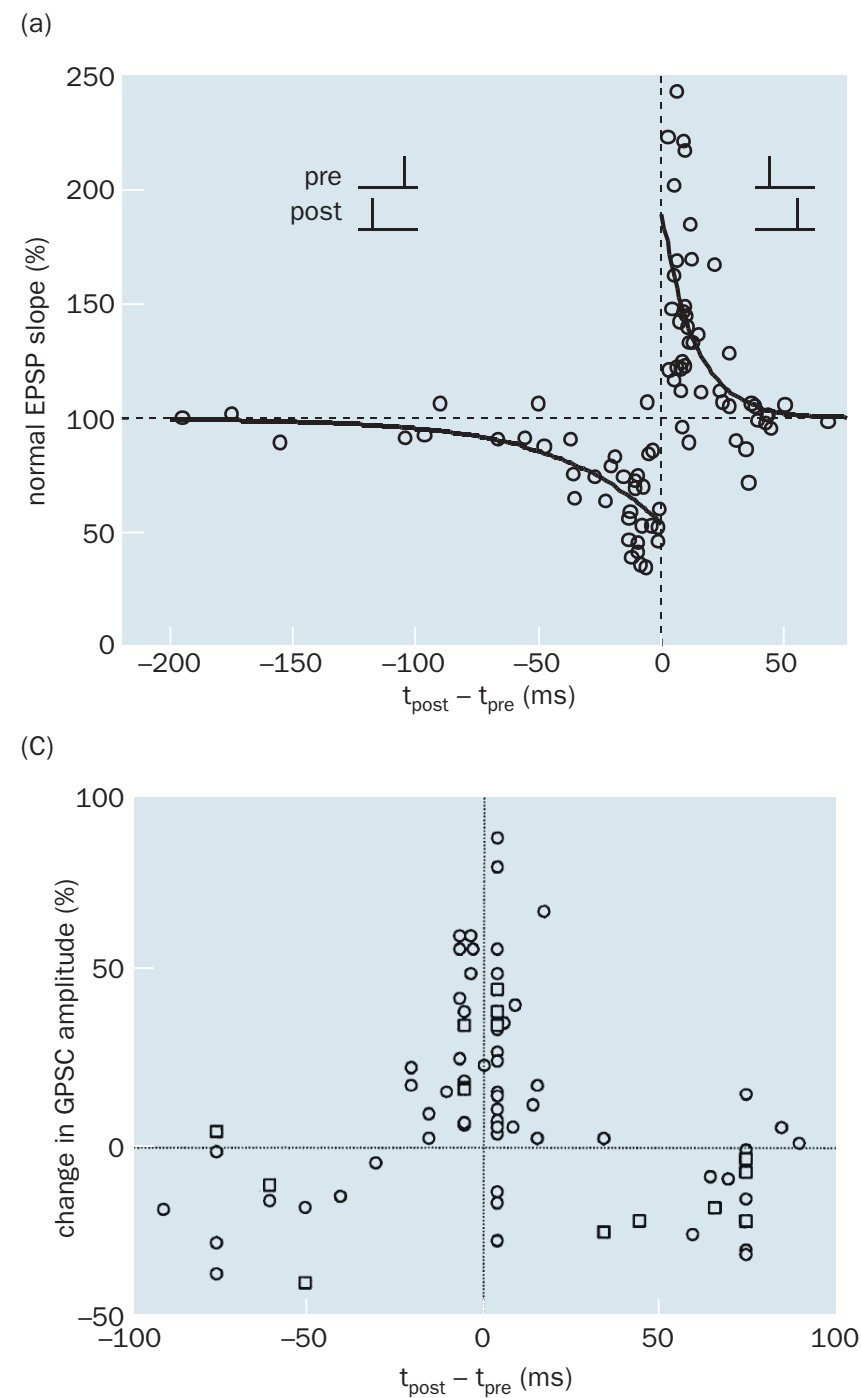
- Local rule — only uses information available at the synapse
- Unsupervised — no external error signal required
- Unstable — weights grow without bound (needs normalization)

Spike-timing-dependent plasticity



Markram et. al., 1997

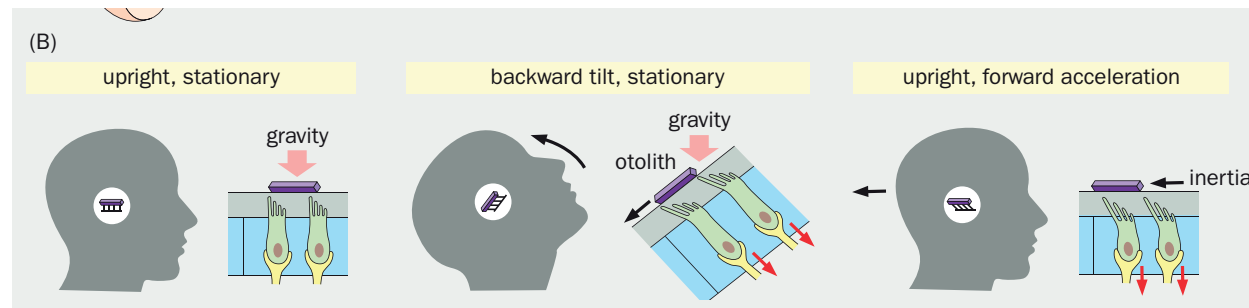
Spike-timing-dependent plasticity



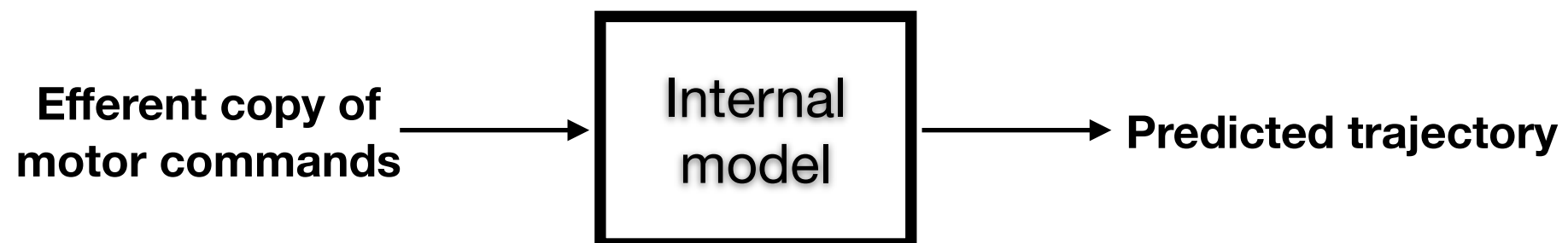
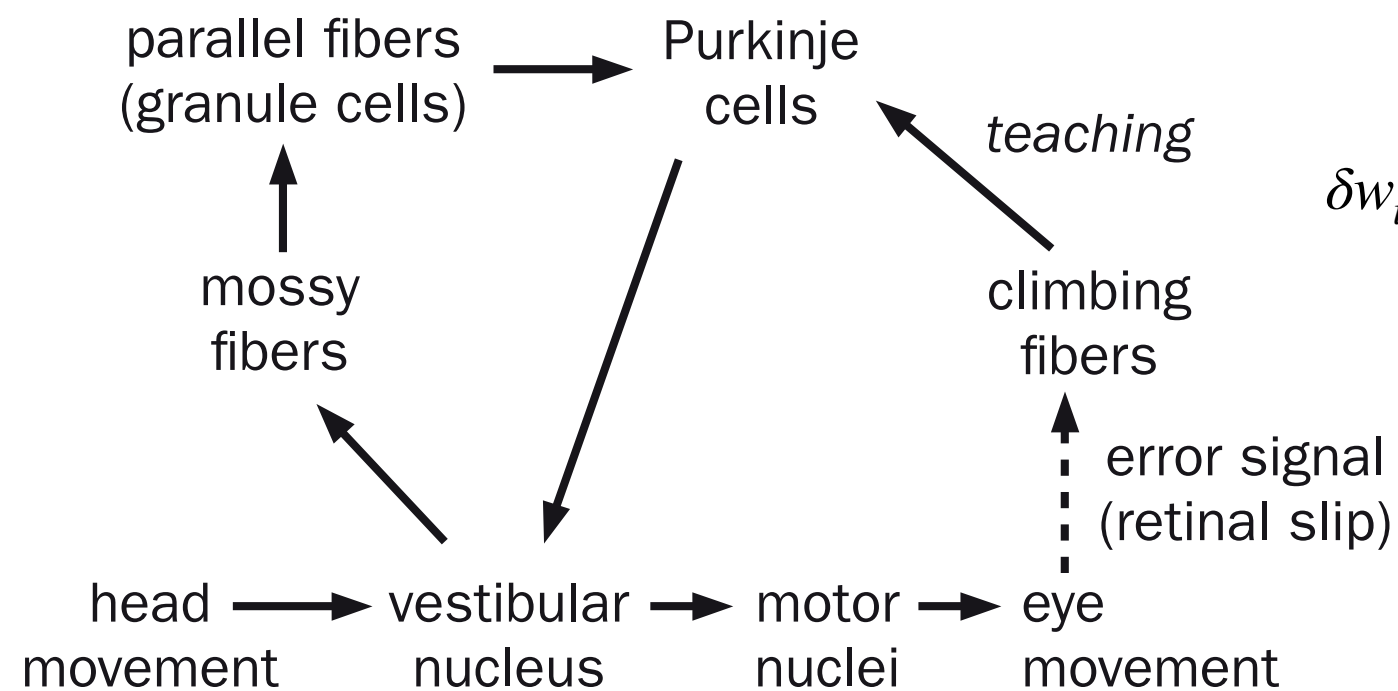
Bi & Poo, 1999, 2001, and others

Supervised learning in the cerebellum

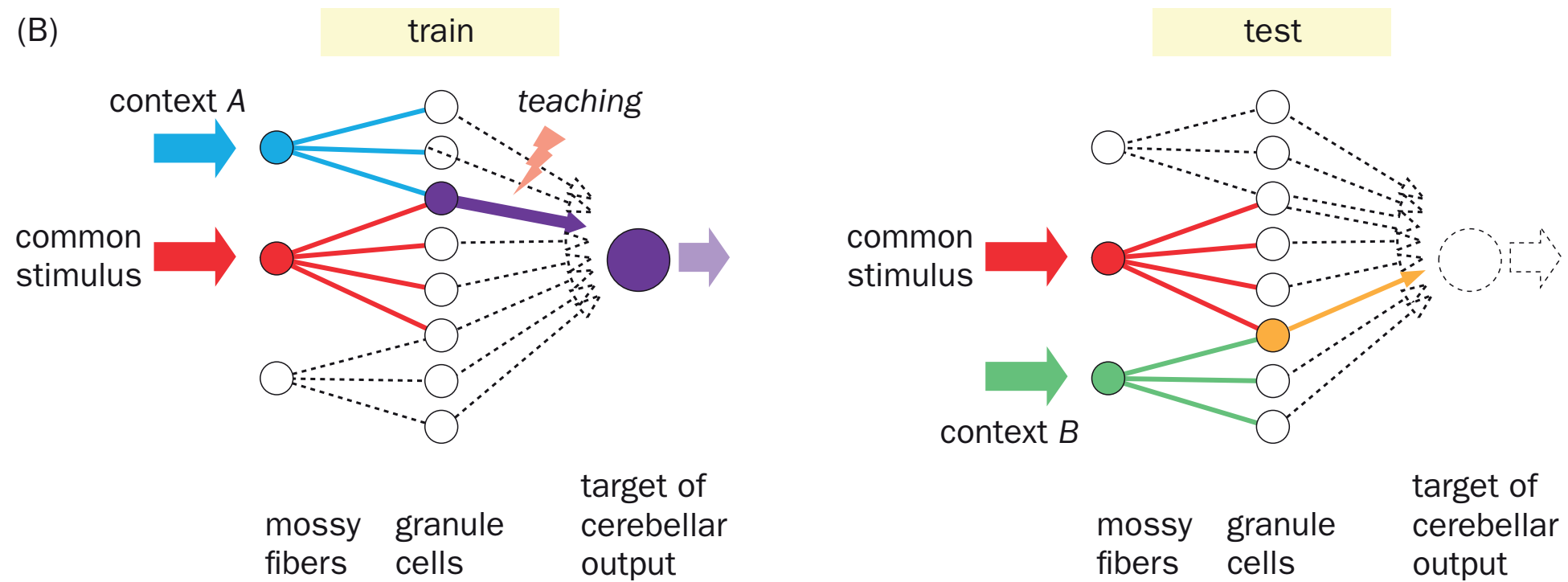
Cerebellar function in Vestibulo-ocular reflex



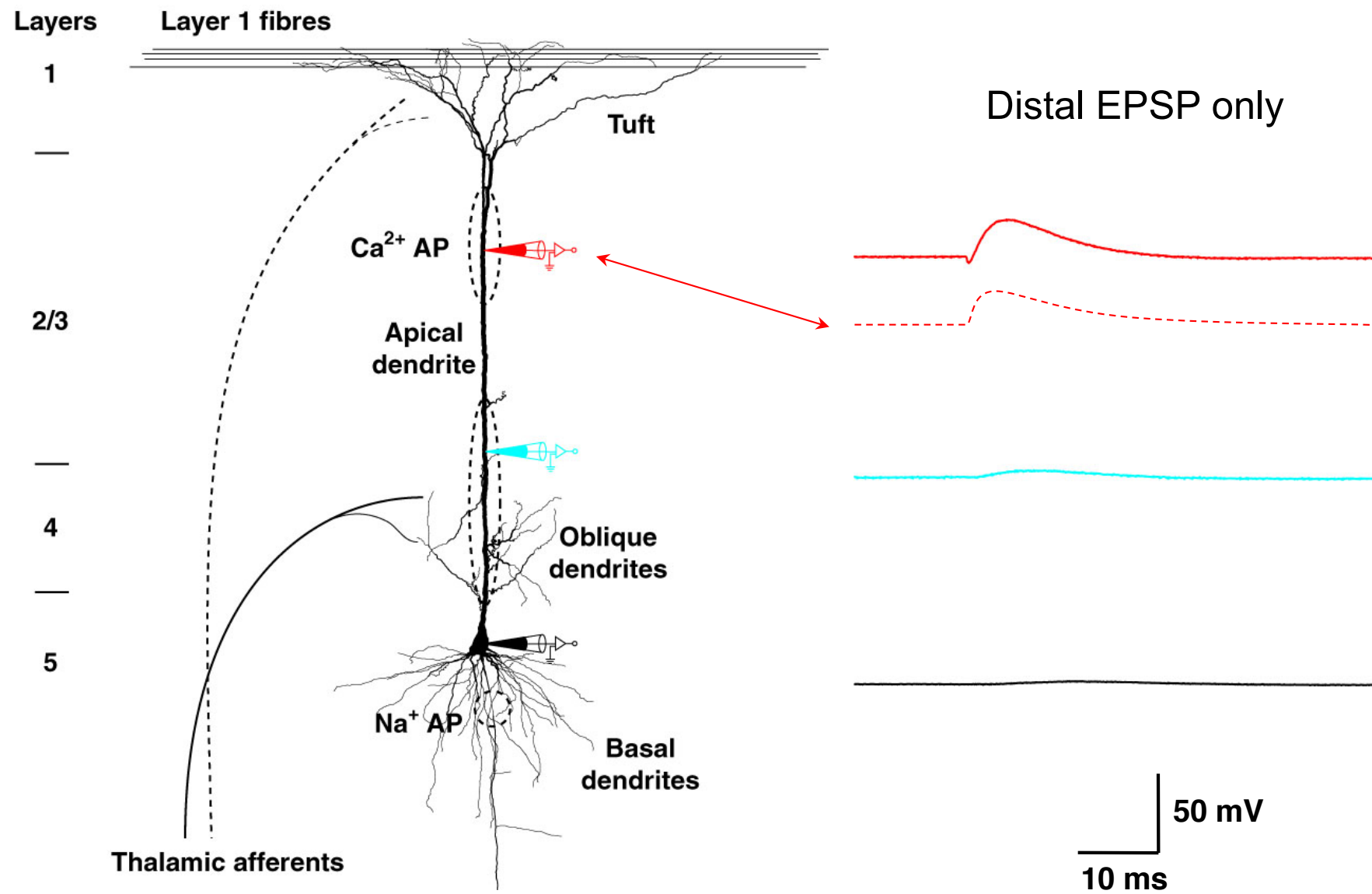
(A)



Supervised learning in the cerebellum

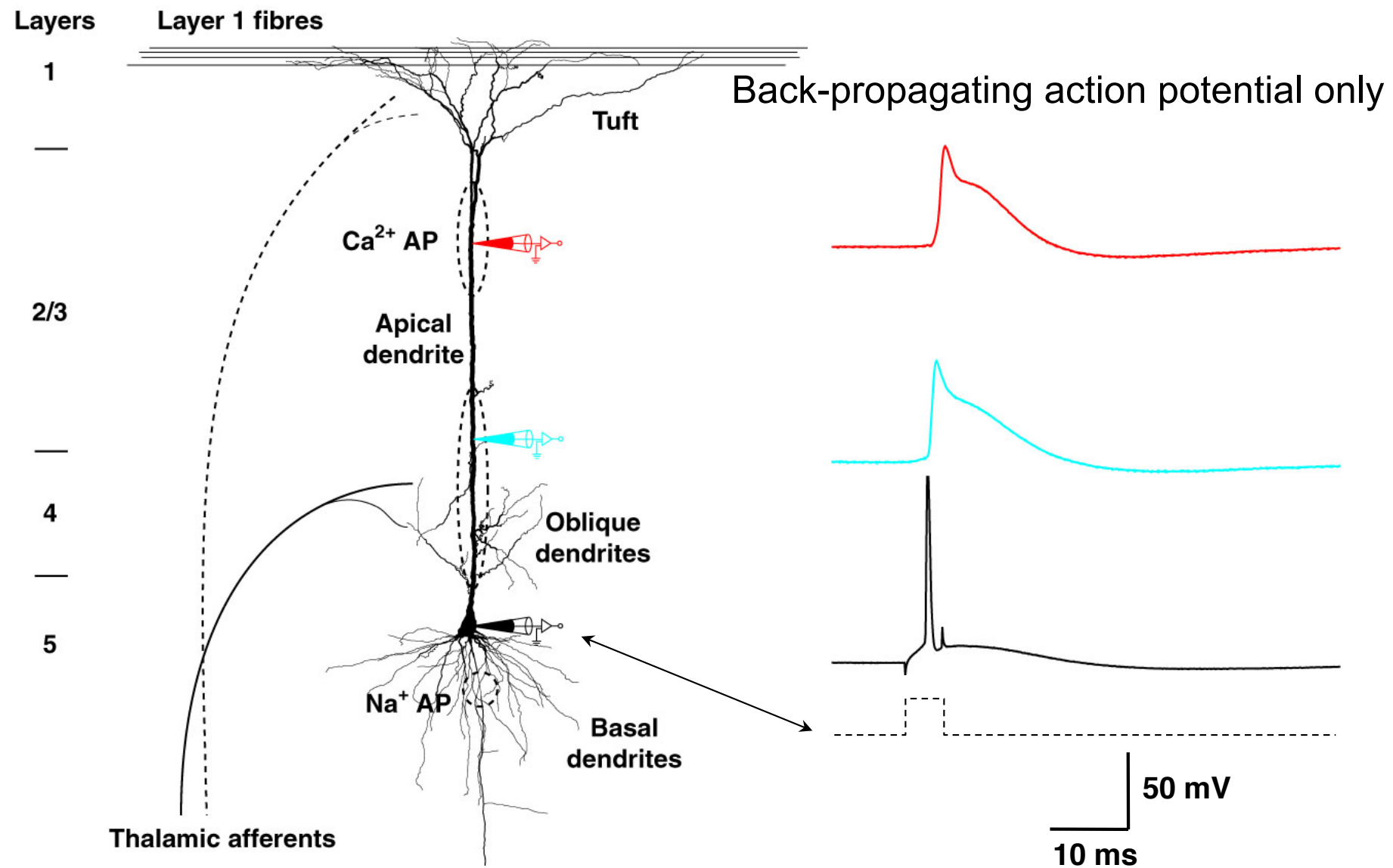


Dendritic Ca^{2+} plateau potential



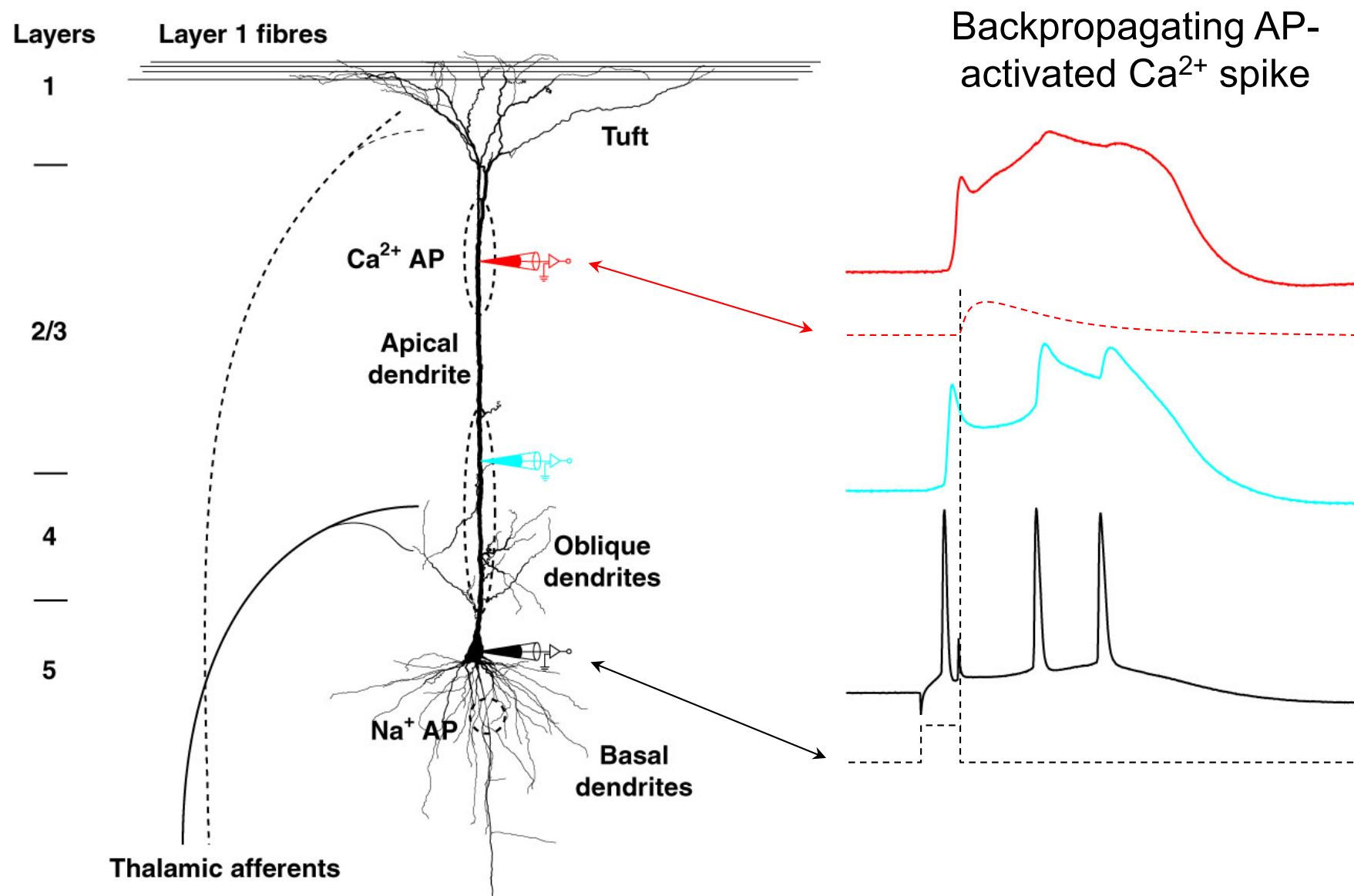
Larkum, Zhu & Sakmann - Nature, 1999

Dendritic Ca^{2+} plateau potential



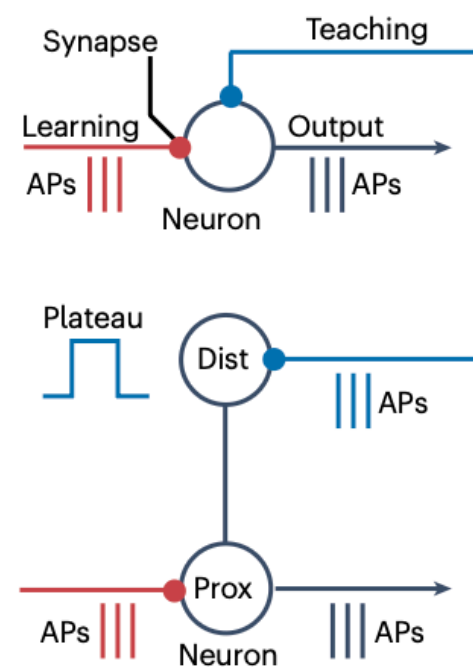
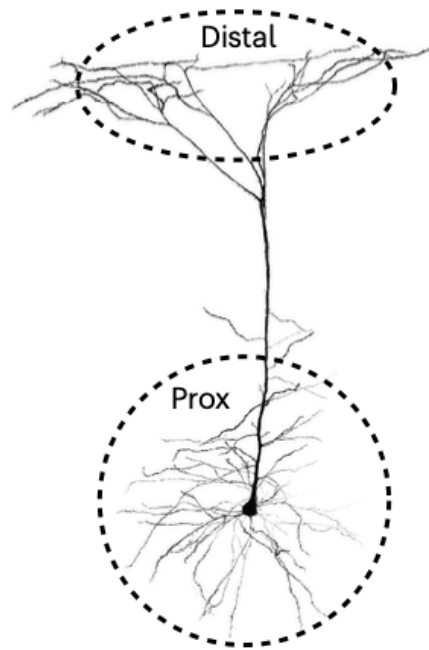
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Dendritic Ca^{2+} plateau potential

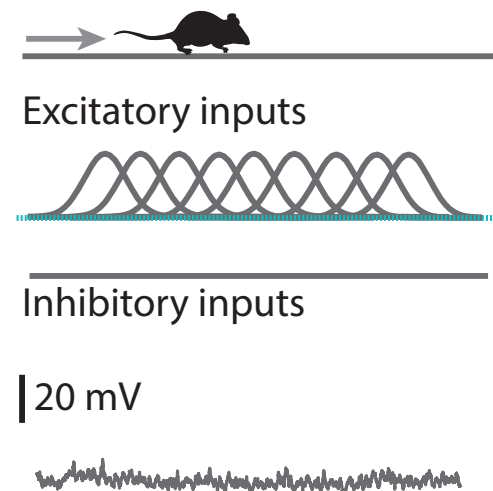


Larkum, Zhu & Sakmann - Nature, 1999

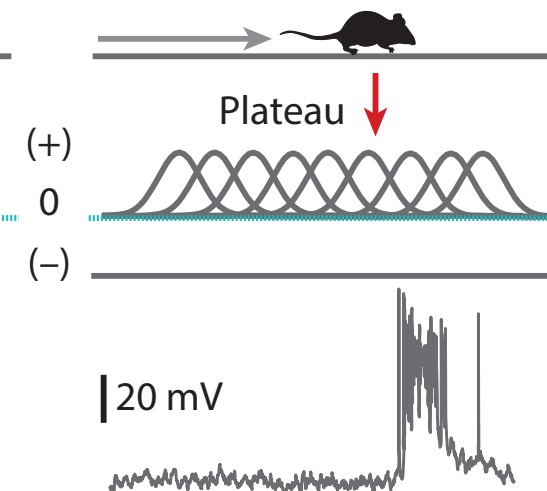
Behavior time-scale plasticity (BTSP) in Hippocampus



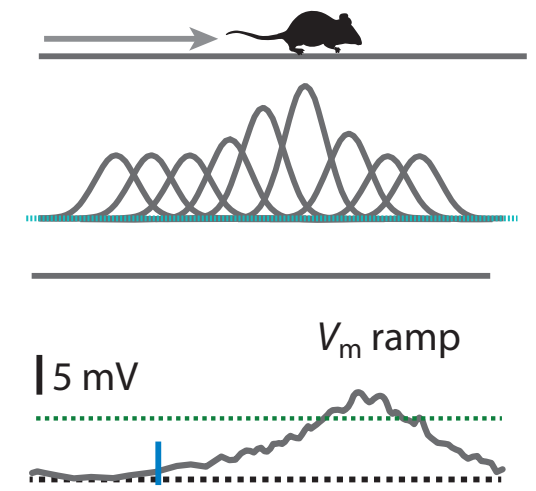
a Silent cell



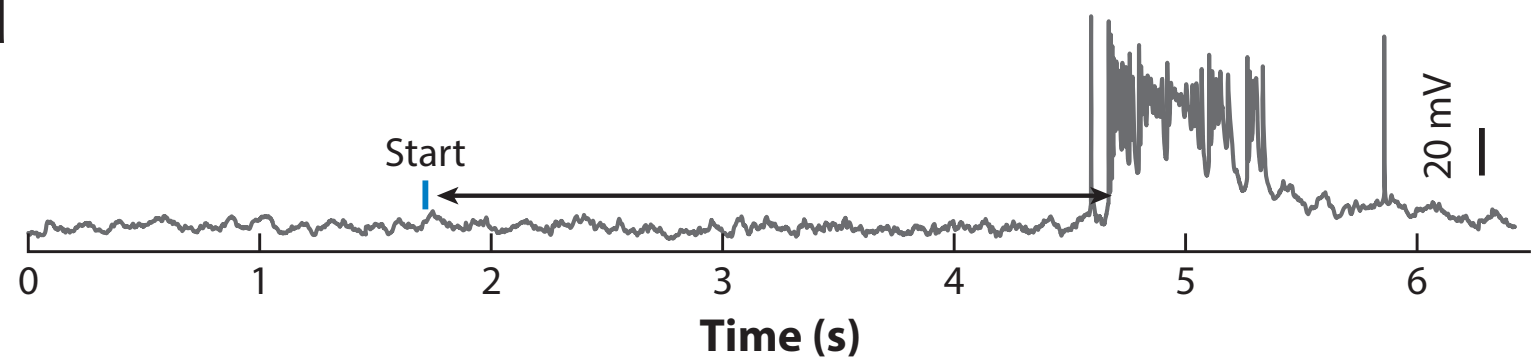
b Plasticity induction



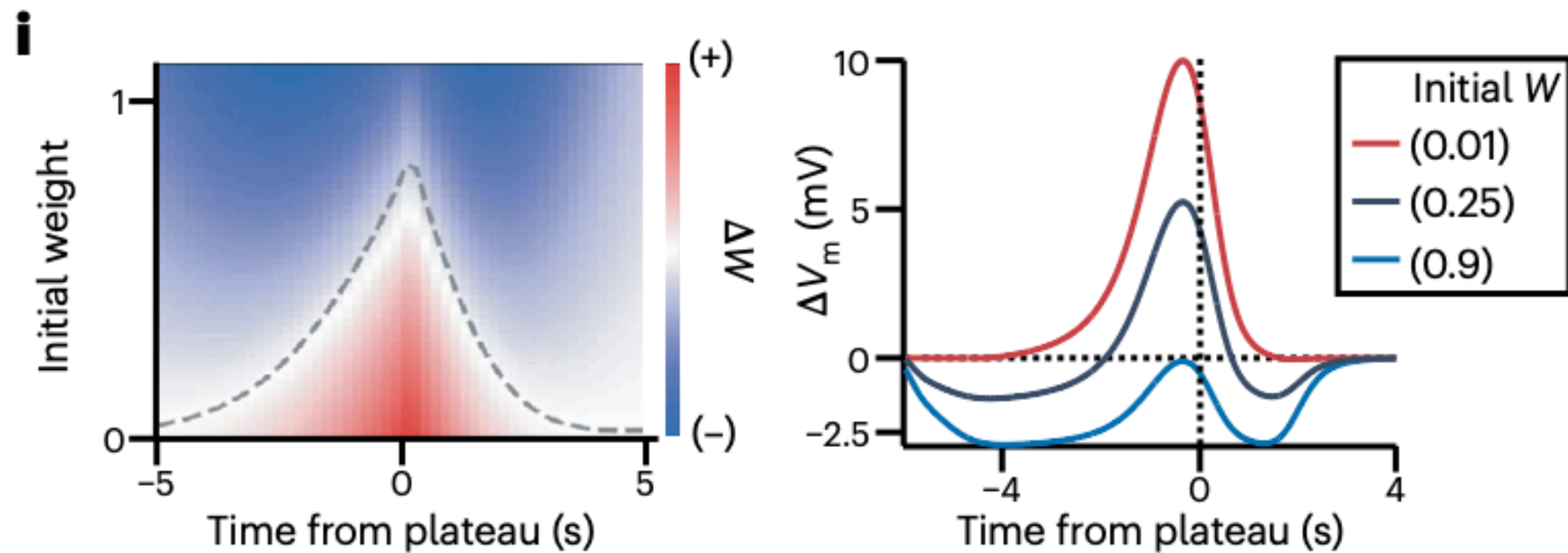
c Place cell



d

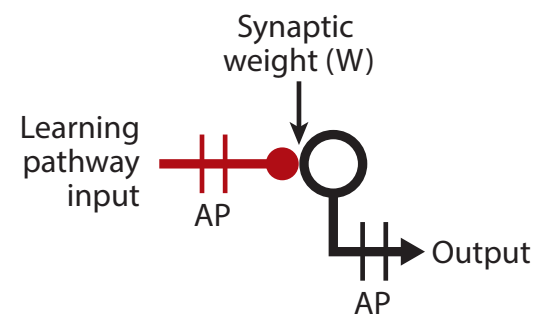


Behavior time-scale plasticity (BTSP) in Hippocampus



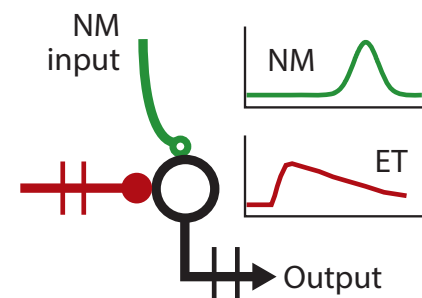
Synaptic learning rule (algorithms)

a Hebbian STDP



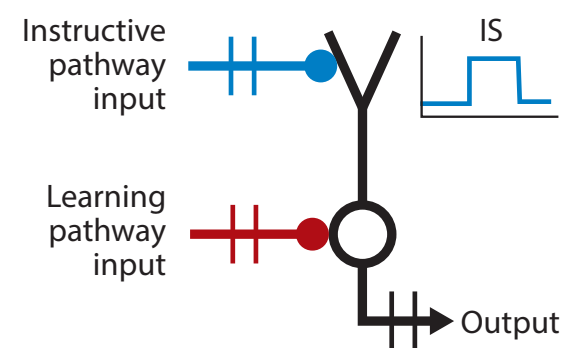
$$\Delta W = \text{input} \times \text{output}$$

b Neuromodulated STDP



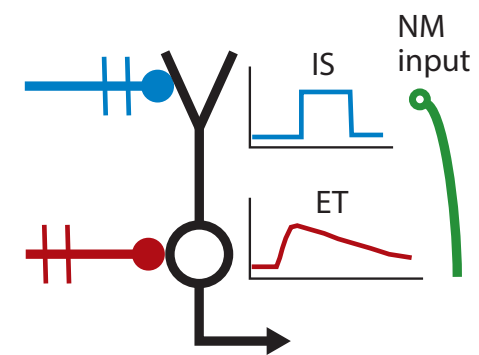
$$\begin{aligned} \text{NM} &= \text{novelty, reward, etc.} \\ \text{ET}_{(t+1)} &= (\text{input} \times \text{output}) - \text{ET}_{(t)} / \tau_e \\ \Delta W &= \text{ET} \times \text{NM} \end{aligned}$$

c Supervised



$$\begin{aligned} \text{IS} &= \text{desired} - \text{actual} \\ \Delta W &= \text{input} \times \text{IS} \end{aligned}$$

d BTSP



$$\begin{aligned} \text{IS} &= \text{local error} \cdot \text{NM} \\ \text{ET}_{(t+1)} &= \text{input} - \text{ET}_{(t)} / \tau_e \\ \Delta W &= \text{ET} \times \text{IS} \end{aligned}$$

NM: neuromodulator

ET: eligibility trace (contingent vs non-contingent)

IS: instructive signal

Rich vs single learning rule

Table 1. The most commonly studied forms of plasticity all appear to be relevant to gradient estimation	
Cell mechanisms	Relation to gradients
Hebbian plasticity	Gradient is zero for inactive cells
STDP	Gradient is zero if presynaptic spikes are too late
Probabilistic release	If release affects performance, gradient is non-zero
No action potential propagation into hyperpolarized segments	Gradient for synapses on a dendrite will be zero if a local dendrite prevents action potential propagation into its segments
CamKII	Recent concurrent pre- and postsynaptic activity indicates potential gradient signal
High-frequency bursts	Bursts can communicate strong relevance of a neuron to performance gradient
Circuit mechanisms	Relation to gradients
Dopamine	Gradients are large if we misestimate reward; dopamine can carry nature or error
Acetylcholine	Gradients should be zero if the situation is irrelevant
Intercellular messages (e.g. NO, RNA, peptides)	Neurons can communicate their contribution to gradients to other neurons using a variety of intercellular messenger systems
Climbing fibres in the cerebellum	Learning should be enabled by errors and can be used to estimate gradients
Attention	Only those parts of the brain that matter for the decision contribute to gradients

Richards & Kording 2023

In different contexts, using different neurons to approximate gradients

$$w_{ji} \leftarrow w_{ji} - \eta \delta_j^l r_i^{l-1}$$

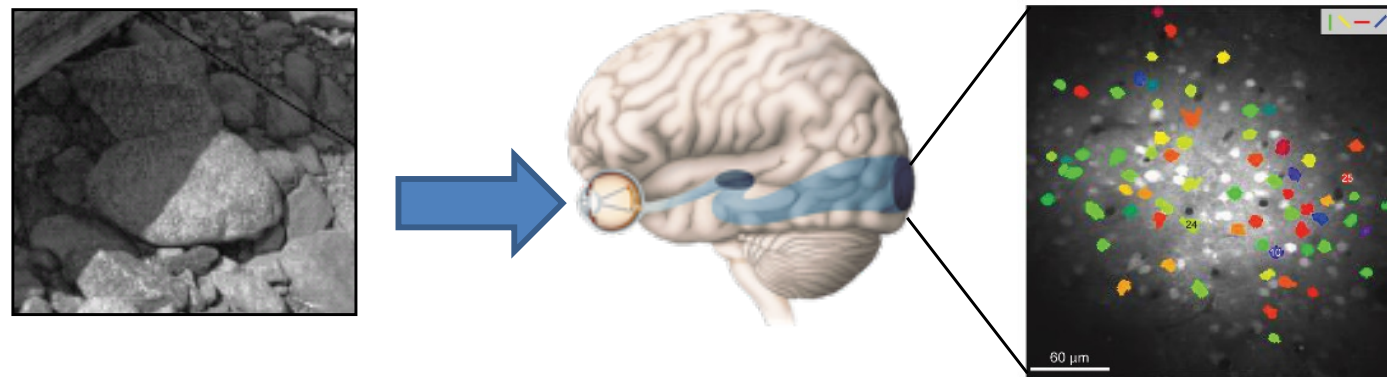
$$\delta_j^l = F'(I_j^l) \sum_k \delta_k^{l+1} w_{kj}$$

Back-propagation

Hebbian Learning vs. Backpropagation

Feature	Hebbian / STDP	Backpropagation
Error Signal	?	Global loss gradient
Locality	Fully local	Non-local (weight transport)
Learning Type	?	Error-driven
Credit Assignment	?	Exact (chain rule)
Biological Plausibility	High	Low
Deep Network Training	Ineffective alone	Highly effective
Stability	Unstable (unbounded)	Stable (with regularization)

Reconstruction approach: linear decoding

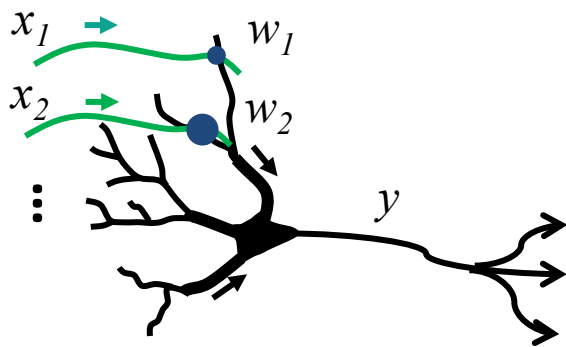


$$\mathbf{x}(t) \approx \mathbf{w}_1 y_1 + \mathbf{w}_2 y_2 + \dots + \mathbf{w}_k y_k$$

$\mathbf{w}_{1:k}$

A 4x4 grid of 16 small grayscale images, each showing a different pattern of neural activity. These patterns represent the weights $\mathbf{w}_1$ through $\mathbf{w}_k$ used in the linear decoding equation.

Reconstructing input signals from single neuron

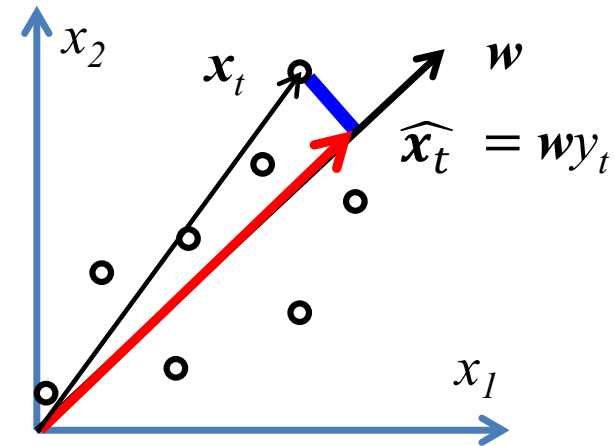


Encoding:

$$y_t = \mathbf{w}^T \mathbf{x}_t, \|\mathbf{w}\|_2^2 = 1$$

Decoding:

$$\hat{\mathbf{x}}_t = \mathbf{w} y_t$$



$$\min_{\hat{\mathbf{x}}_t} \sum_{t=1}^T \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|_2^2$$

$$\min_{\substack{\mathbf{w}, y_t \\ \|\mathbf{w}\|=1}} \sum_{t=1}^T \|\mathbf{x}_t - \mathbf{w} y_t\|_2^2$$

$$\operatorname{argmin}_{\mathbf{w}, \|\mathbf{w}\|=1} \sum_{t=1}^T \|\mathbf{x}_t - \mathbf{w} \mathbf{w}^T \mathbf{x}_t\|_2^2 = \operatorname{argmax}_{\mathbf{w}, \|\mathbf{w}\|=1} \sum_{t=1}^T \mathbf{w}^T \mathbf{x}_t \mathbf{x}_t^T \mathbf{w}$$

$$= \operatorname{argmax}_{\mathbf{w}, \|\mathbf{w}\|=1} \mathbf{w}^T \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^T \mathbf{w} = \operatorname{eig}_1 \left(\sum_t \mathbf{x}_t \mathbf{x}_t^T \right)$$

Online dimensionality reduction top principal component

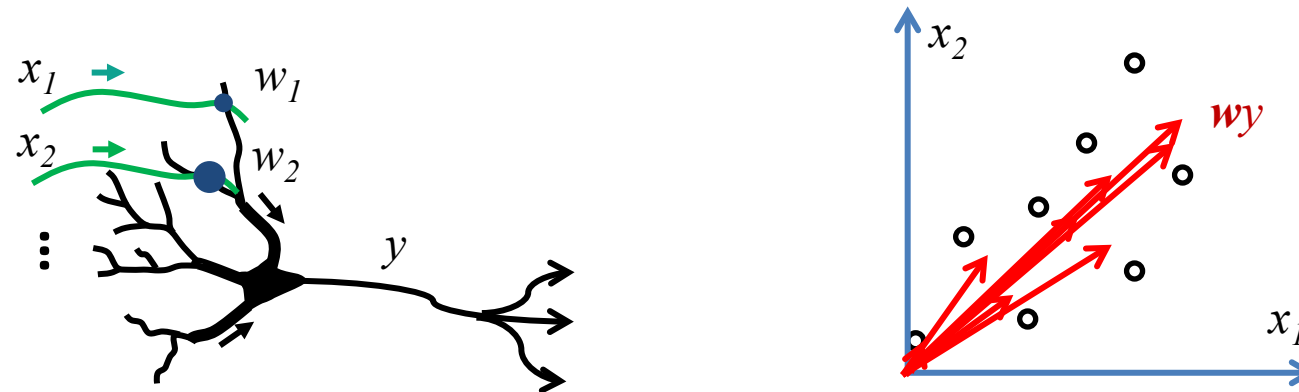


$$\min_{\mathbf{w}, \|\mathbf{w}\|=1} \sum_{t=1}^T \min_{y_t} \|\mathbf{x}_t - \mathbf{w}y_t\|_2^2$$

Phase 1: minimize neural activity

$$y_t = \mathbf{w}_{t-1}^T \mathbf{x}_t$$

Online dimensionality reduction top principal component

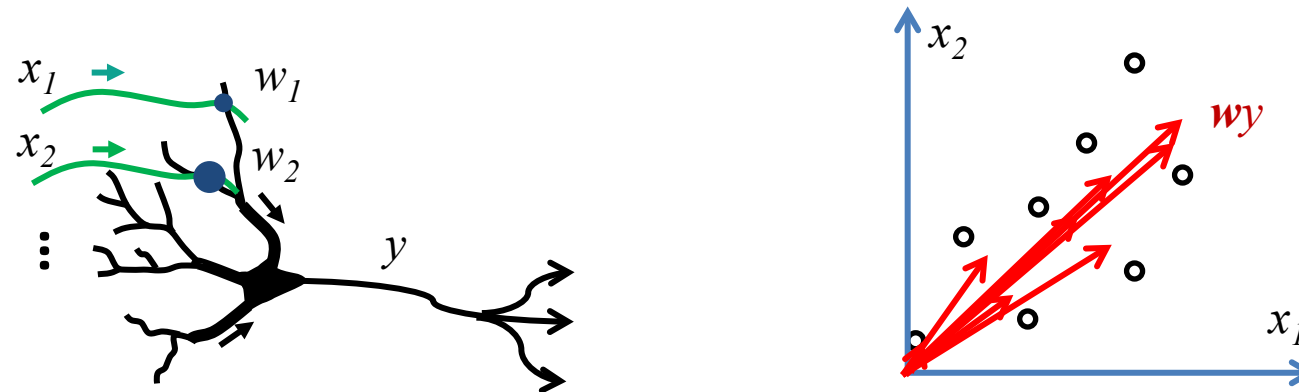


$$\min_{\mathbf{w}, \|\mathbf{w}\|=1} \sum_{t=1}^T \min_{y_t} \|\mathbf{x}_t - \mathbf{w}y_t\|_2^2$$

Phase 2: minimize synaptic weight

$$\begin{aligned} \mathbf{w}_T &= \arg \min_{\mathbf{w}} \sum_{t=1}^T \|\mathbf{x}_t - \mathbf{w}y_t\|_2^2 \\ &= \arg \min_{\mathbf{w}} \sum_{t=1}^T \left[-\mathbf{w}^T \mathbf{x}_t y_t + \|\mathbf{w}\|_2^2 y_t^2 \right] = \frac{\sum_{t=1}^T \mathbf{x}_t y_t}{\sum_{t=1}^T y_t^2} \end{aligned}$$

Online dimensionality reduction top principal component



$$\min_{\mathbf{w}, \|\mathbf{w}\|=1} \sum_{t=1}^T \min_{y_t} \|\mathbf{x}_t - \mathbf{w}y_t\|_2^2$$

Phase 2: minimize synaptic weight

$$\mathbf{w}_T = \mathbf{w}_{T-1} + y_T (\mathbf{x}_T - \mathbf{w}_{T-1}y_T) / \sum_{t=1}^T y_t^2$$

Hebbian plasticity and Oja rule

neural activity: $y_t = \mathbf{w}_{t-1}^T \mathbf{x}_t$

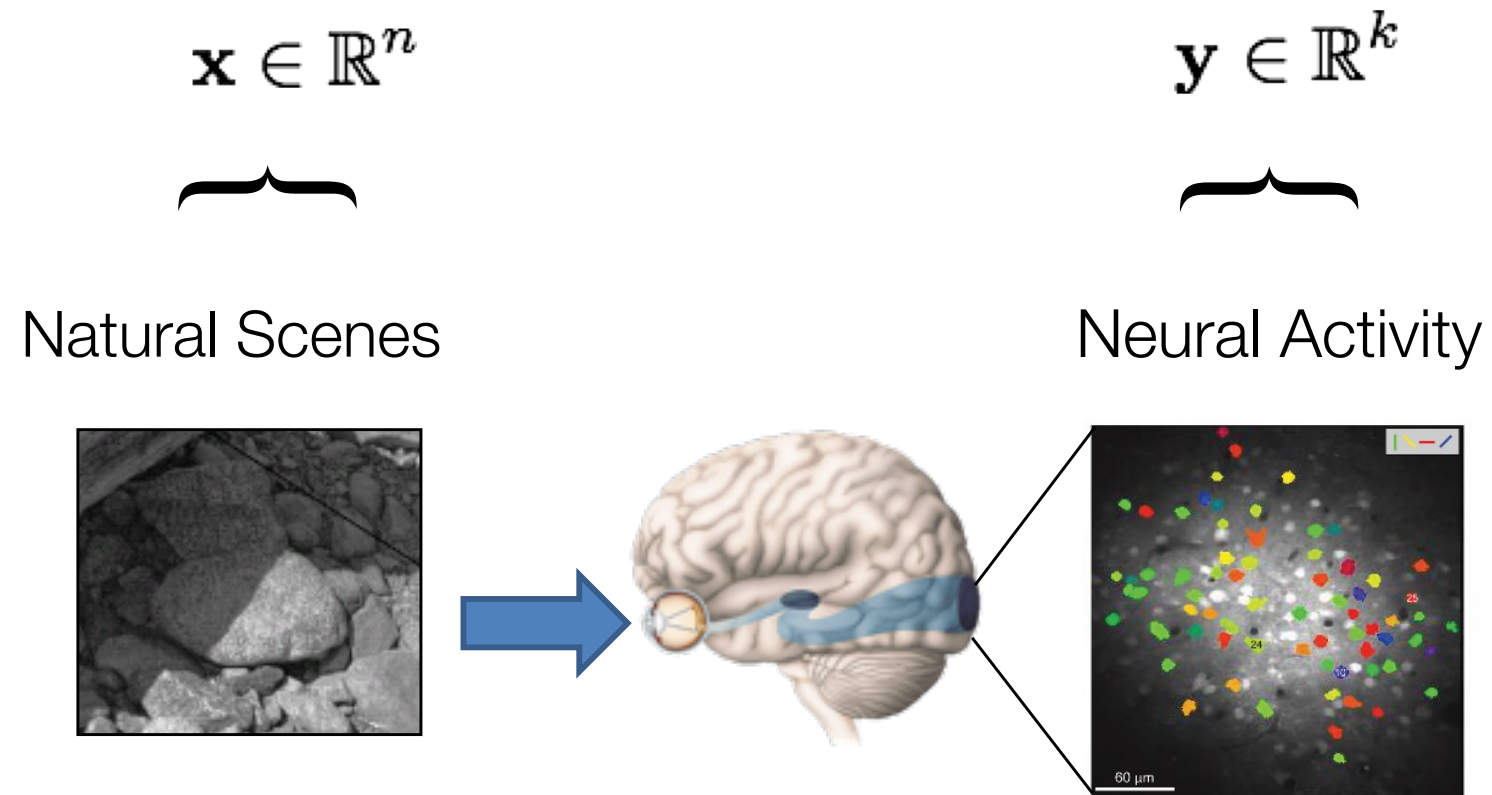
synaptic weight:

$$\mathbf{w}_T = \mathbf{w}_{T-1} + y_T (\mathbf{x}_T - \mathbf{w}_{T-1} y_T) / \sum_{t=1}^T y_t^2$$

$$\tau_w \frac{d\mathbf{w}}{dt} = y\mathbf{x} - y^2\mathbf{w},$$

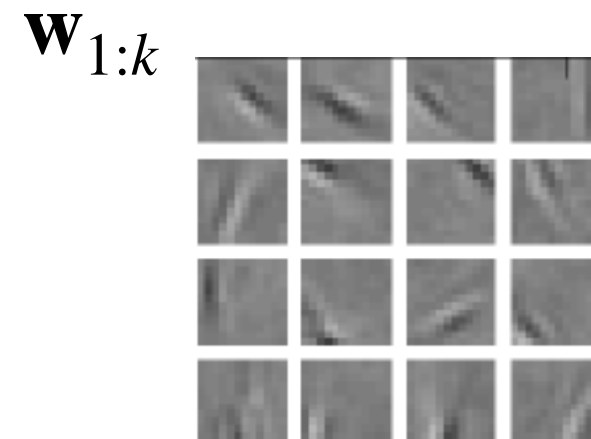
$$\tau_w \frac{d\|\mathbf{w}\|^2}{dt} = 2\tau_w \mathbf{w} \cdot \frac{d\mathbf{w}}{dt} = 2y^2(1 - \|\mathbf{w}\|^2)$$

synaptic weight is bounded!



$$\mathbf{x}(t) \approx \mathbf{w}_1 y_1(t) + \mathbf{w}_2 y_2(t) + \dots + \mathbf{w}_k y_k(t)$$

$$\mathbf{x}(t) \approx \mathbf{W} \mathbf{y}(t)$$



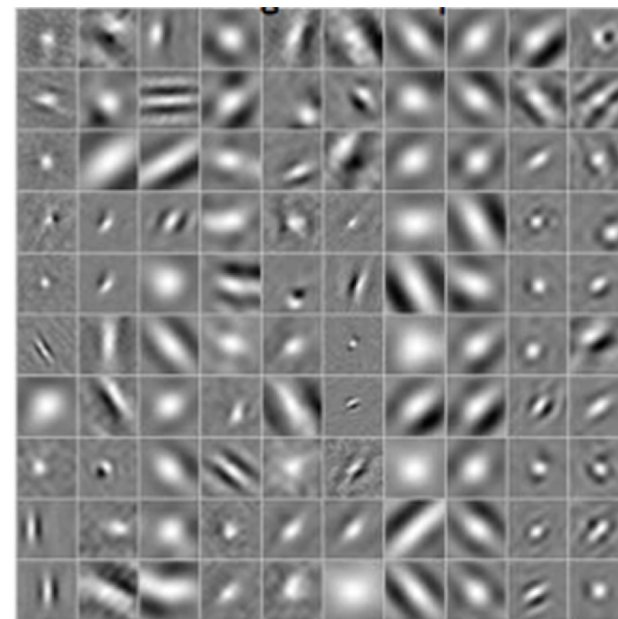
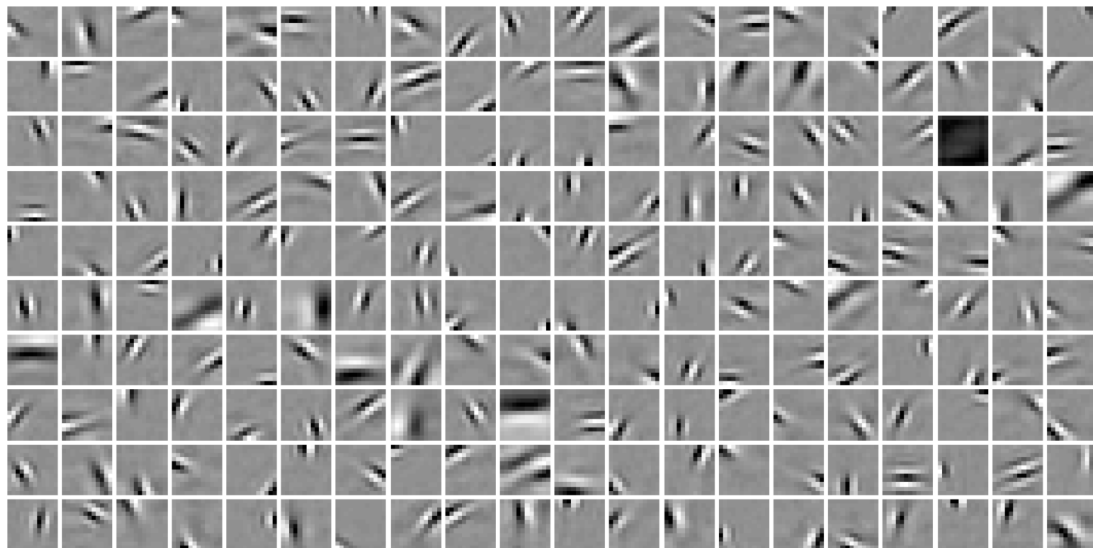
Efficient reconstruction theory:

$$\min_{\mathbf{W}} \sum_t \min_{\mathbf{y}_t} \|\mathbf{x}_t - \mathbf{W} \mathbf{y}_t\|_2^2 + \text{constraints}$$

Sparse coding

Olshausen & Field (1996): Sparse, overcomplete ($k > n$) representations

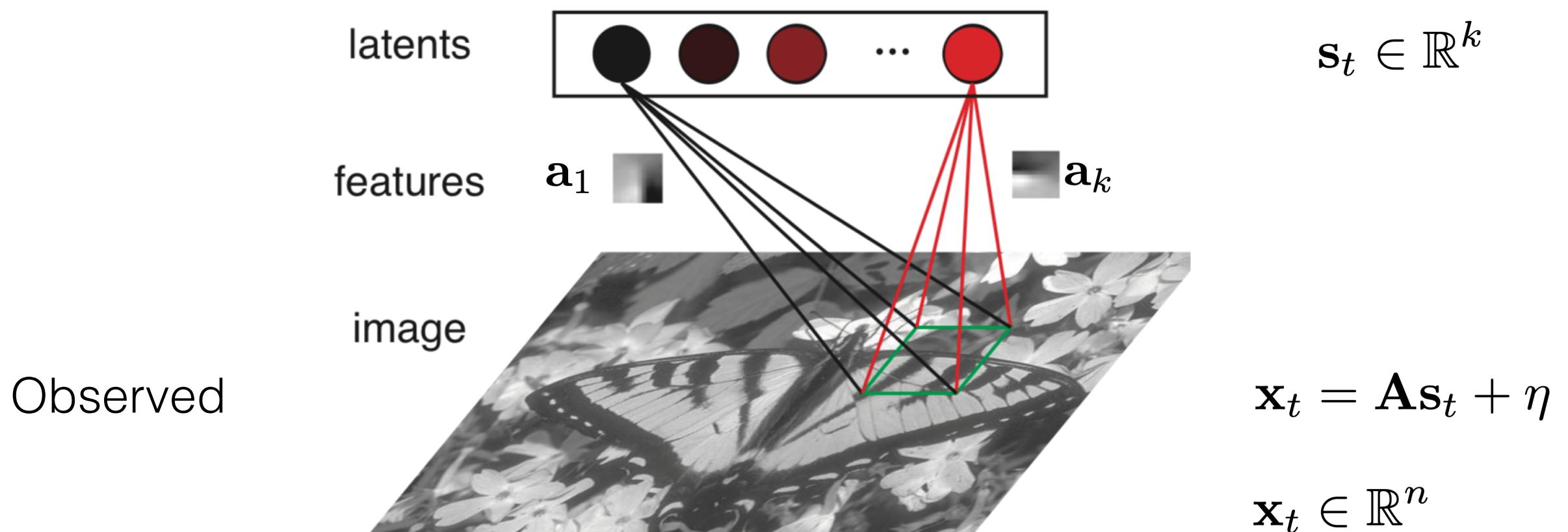
$$\min_{\mathbf{W}} \sum_t \min_{\mathbf{y}_t} \|\mathbf{x}_t - \mathbf{W}\mathbf{y}_t\|_2^2 + \lambda \sum_t \|\mathbf{y}_t\|_1$$



*Data from
Ringach lab*

Probabilistic interpretation of Sparse Coding

Sparse coding is based on a “generative model”. Task is to recover “latent” generating factors.



How do we infer the latent factors from the observed variables? For $n < k$, this is a terribly ill-defined problem. Need to assume something! Priors!

Probabilistic interpretation of Sparse Coding

$$\mathbf{x}_t = \mathbf{A}\mathbf{s}_t + \eta \quad \mathbf{x}_t \in \mathbb{R}^n, \quad \mathbf{s}_t \in \mathbb{R}^k$$

$$\text{Prior: } p(\mathbf{s}_t) \propto e^{-\lambda \|\mathbf{s}_t\|_1} \quad \text{Conditional: } p(\mathbf{x}_t | \mathbf{s}_t; \mathbf{A}) \propto e^{-\frac{\|\mathbf{x}_t - \mathbf{A}\mathbf{s}_t\|_2^2}{2\sigma_\eta^2}}$$

$$\text{Model likelihood: } p(\mathbf{x}_t; \mathbf{A}) = \int d\mathbf{s}_t p(\mathbf{x}_t | \mathbf{s}_t; \mathbf{A}) p(\mathbf{s}_t)$$

$$\begin{aligned} \max_{\mathbf{A}} \log p(\mathbf{x}_t; \mathbf{A}) &= \max_{\mathbf{A}} \log \int d\mathbf{s}_t p(\mathbf{x}_t | \mathbf{s}_t; \mathbf{A}) p(\mathbf{s}_t) \\ &\approx \max_{\mathbf{A}} \log \max_{\mathbf{s}_t} p(\mathbf{x}_t | \mathbf{s}_t; \mathbf{A}) p(\mathbf{s}_t) \\ &= \max_{\mathbf{A}} \max_{\mathbf{s}_t} \log p(\mathbf{x}_t | \mathbf{s}_t; \mathbf{A}) p(\mathbf{s}_t) \\ &= \max_{\mathbf{A}} \max_{\mathbf{s}_t} -\frac{\|\mathbf{x}_t - \mathbf{A}\mathbf{s}_t\|_2^2}{2\sigma_n^2} - \lambda \|\mathbf{s}_t\|_1 + \dots \end{aligned}$$

$$\mathbf{x}_t = \mathbf{A}\mathbf{s}_t + \eta \qquad \mathbf{x}_t \in \mathbb{R}^n, \quad \mathbf{s}_t \in \mathbb{R}^k$$

$$\max_{\mathbf{A}} \max_{\mathbf{s}_t} - \frac{\|\mathbf{x}_t - \mathbf{A}\mathbf{s}_t\|_2^2}{2\sigma_n^2} - \lambda \|\mathbf{s}_t\|_1$$

Infer latent
Variables
(Neural activity)

Update
Generative
Model
(Plasticity)

The cost function

$$E = \|\mathbf{x} - \mathbf{W}\mathbf{y}\|_2^2 + \lambda_2 g(\mathbf{y}) + \lambda \sum_{i,j} W_{ij}^2$$

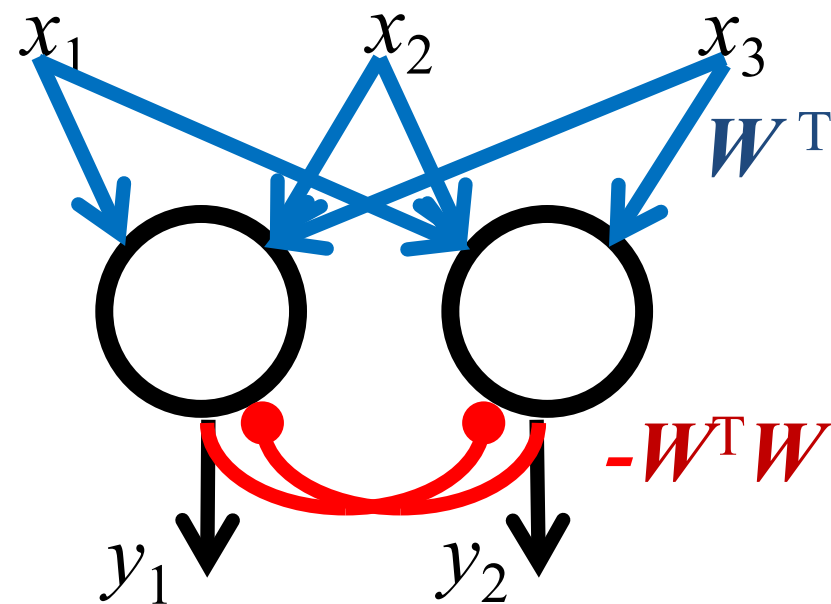
$$g(\mathbf{y}) = \|\mathbf{y}\|_2^2 \text{ gaussian prior}$$

$$= \sum_i \log(1 + y_i^2) \text{ sparse prior}$$

neural activity: $\frac{d\mathbf{y}}{dt} = -\frac{\partial E}{\partial \mathbf{y}}$

synaptic weight: $\frac{d\mathbf{W}}{dt} = -\frac{\partial E}{\partial \mathbf{W}}$

Neural circuit implementation (single layer)



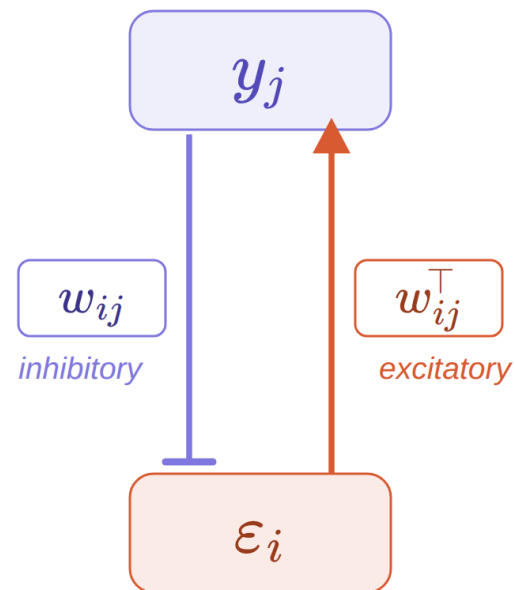
$$\frac{d\mathbf{y}}{dt} = -\lambda_2 \mathbf{y} + W^T \mathbf{x} - W^T W \mathbf{y}$$

HW: Derive this!

$$\frac{dW_{ij}}{dt} = -\lambda W_{ij} + x_i y_j - y_j \sum_k W_{ik} y_k \quad \text{Non-local learning rule!}$$

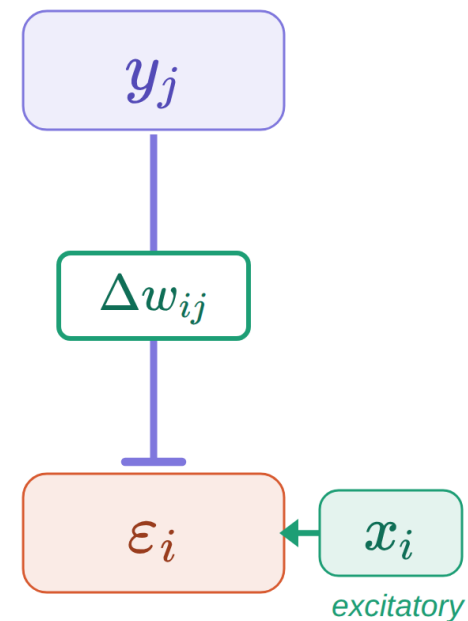
Predictive coding (two layers)

Phase 1 — Inference



Same weight used in both directions — $\triangle$
weight transport problem

Phase 2 — Weight update



Local update: postsynaptic $\varepsilon_i \times$
presynaptic y_j

▲ Excitatory — Inhibitory

Inference dynamics

$$\varepsilon_i = x_i - \sum_k w_{ik} y_k$$

$$\dot{y}_j = \sum_i w_{ij} \varepsilon_i - y_j$$

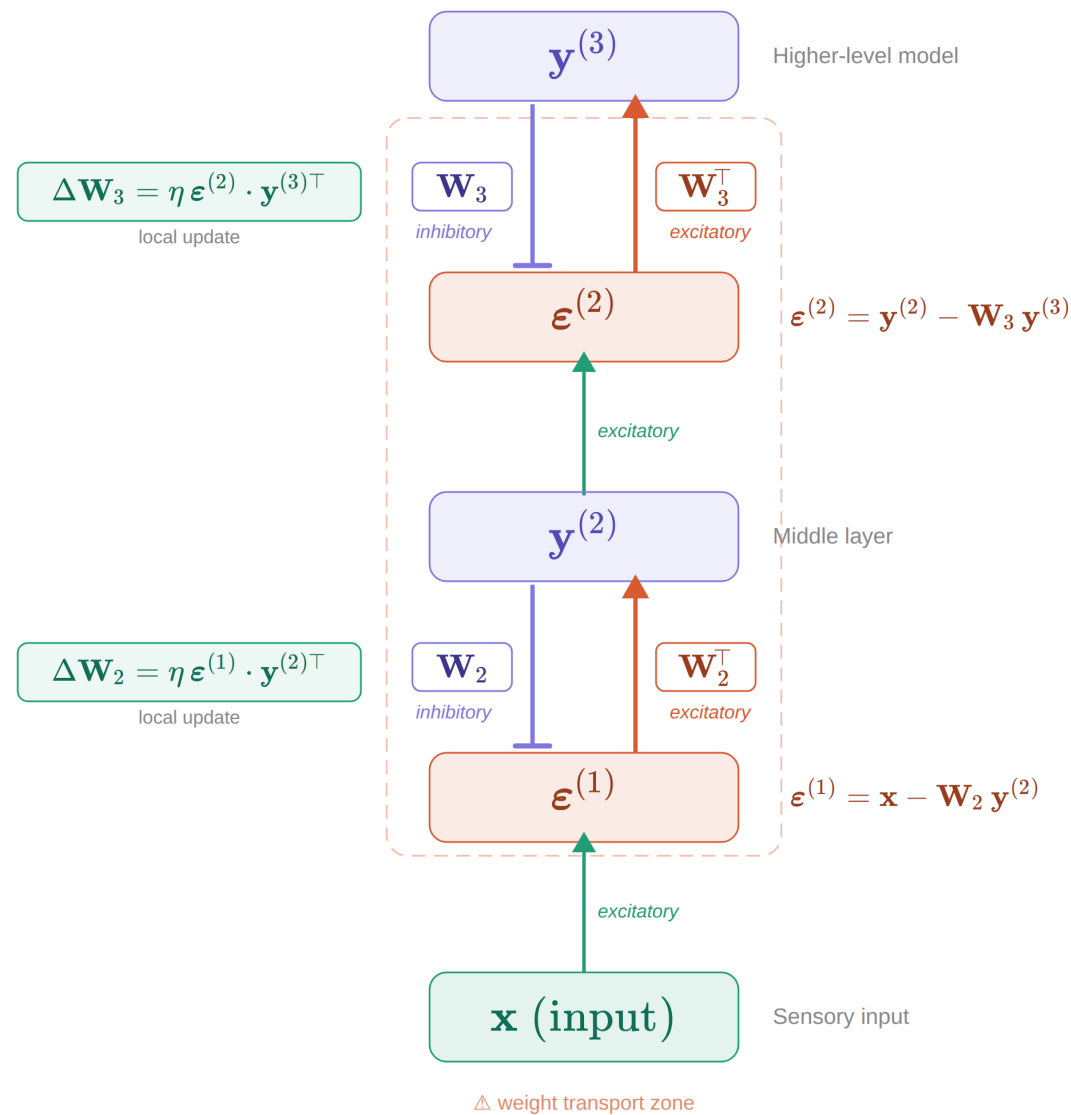
Error propagates back up via $\mathbf{W}^\top$ — requires weight transport

Synaptic plasticity

$$\Delta w_{ij} = \eta \cdot \varepsilon_i \cdot y_j$$

Purely local: postsynaptic $\varepsilon_i \times$ presynaptic y_j

Hierarchical predictive coding



Inference (Phase 1)

$$\epsilon^{(\ell)} = y^{(\ell)} - W_{\ell+1} y^{(\ell+1)}$$

$$\dot{y}^{(\ell)} = -\epsilon^{(\ell)} + W_{\ell}^{\top} \epsilon^{(\ell-1)} - y^{(\ell)}$$

Each layer settles by integrating top-down errors via $W_{\ell}^{\top}$

Plasticity (Phase 2)

$$\Delta W_{\ell} = \eta \cdot \epsilon^{(\ell-1)} \cdot y^{(\ell)\top}$$

Each synapse updates from local error $\times$ local activity — no transpose needed

△ Weight transport

The backward error pathways $W_{\ell}^{\top}$ must mirror the forward prediction pathways W_{ℓ} — two separate physical synapses that must remain identical

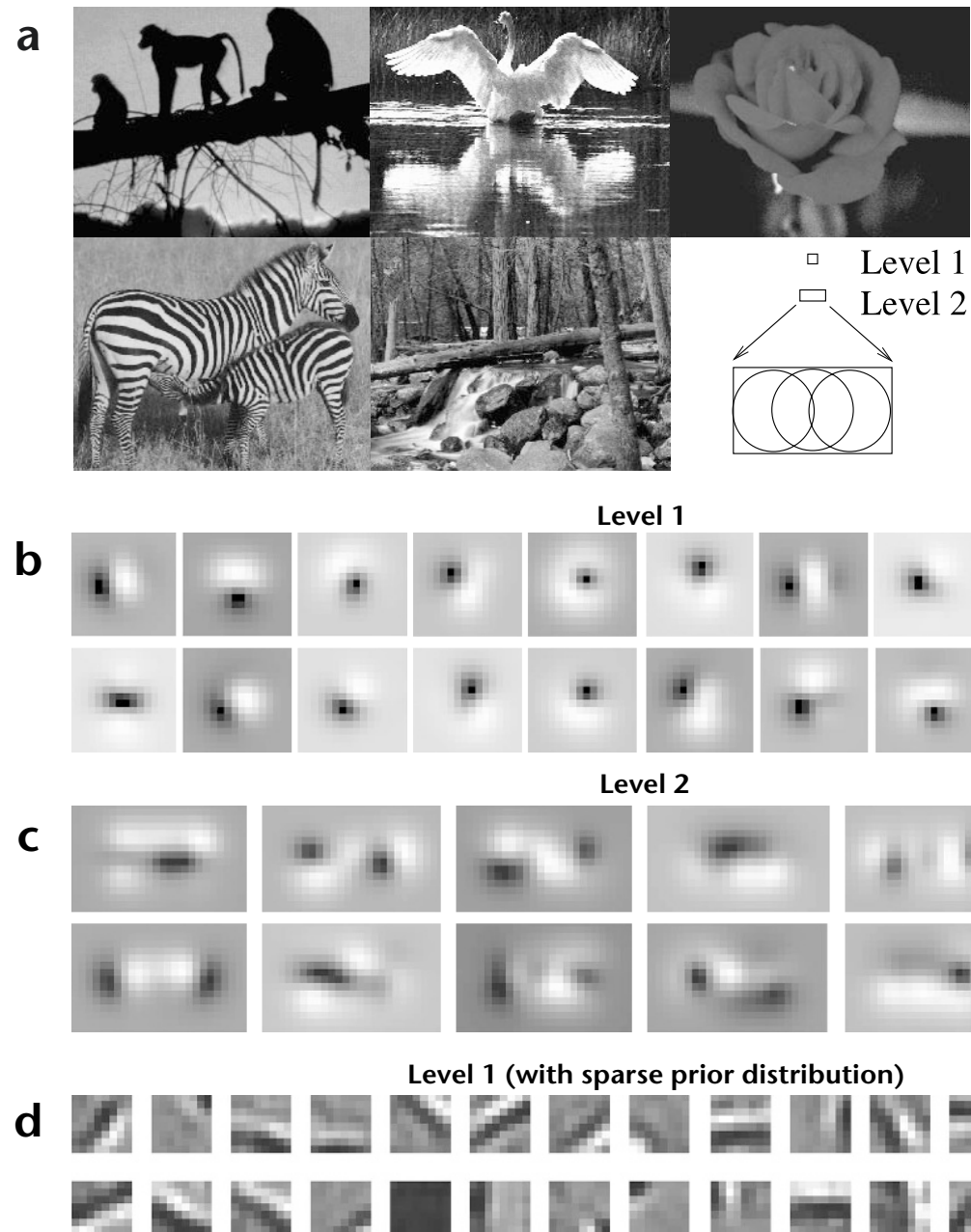


Fig. 2. Receptive fields of feedforward model neurons after training on natural images. **(a)** Five natural images used for training the three-level hierarchical network of Fig. 1c (Methods). The two upper boxes in the bottom right corner show relative sizes (16×16 and 16×26 pixels) of level-1 and level-2 receptive fields respectively. **(b)** Learned synaptic weights (RF weighting profiles) of 20 of the 32 feedforward model neurons in the level-1 module analyzing the central image region. Flanking image regions were analyzed by two other level-1 modules (**Fig. 1c**), each with 32 feedforward model neurons (Methods). Values for these synapses, which form rows of the matrix U^T , can be positive (excitatory, bright regions) or negative (inhibitory, dark regions). These RF profiles resemble classical oriented-edge/bar detectors characteristic of simple cells². **(c)** RF profiles of 12 of the 128 level-2 feedforward model neurons. **(d)** Localized RF profiles resembling Gabor wavelets obtained by using a sigmoidal nonlinearity in the generative model, along with a sparse kurtotic prior distribution for the network activities (Methods). All 32 level-1 feedforward model neurons are shown; Gaussian windowing of inputs (as in b) was not necessary in this case.